

# Stage 3 — Good–Turing Intuition

## 1. Occupancy counts

Suppose histories are treated as categories.

$A, A, A, B, B, C, D$

Then:

$$C(A) = 3, \quad C(B) = 2, \quad C(C) = 1, \quad C(D) = 1$$

Instead of asking:

“How many times did history  $h$  occur?”

ask:

“How many histories occurred exactly  $r$  times?”

Define:

$$f_r = \#\{h : C(h) = r\}$$

Therefore:

$$f_1 = 2, \quad f_2 = 1, \quad f_3 = 1$$

|                                              |
|----------------------------------------------|
| $f_1 = \text{number of singleton histories}$ |
|----------------------------------------------|

## 2. Missing mass

Suppose some histories exist but were never observed in training.

Let:

$$C(h) = 0$$

for an unseen history  $h$ .

The **missing mass** is:

$$M_N = \sum_{h: C(h)=0} p(h)$$

Important:

missing mass  $\neq$  number of unseen histories

Instead:

$$\text{missing mass} = \text{total probability mass assigned to unseen histories}$$

Example:

$$p(E) = 0.05, \quad p(F) = 0.02$$

If both  $E$  and  $F$  are unseen:

$$M_N \supset 0.05 + 0.02 = 0.07$$

### 3. Why singletons tell us something about unseen histories

Suppose:

$$N = 10,000$$

history observations.

Case A:

$$f_1 \approx 0$$

Most observed histories have appeared many times.

$\Rightarrow$  training distribution is relatively well-covered

Case B:

$f_1$  is large

Many histories have appeared only once.

$\Rightarrow$  we are still frequently discovering rare histories

$\Rightarrow$  unseen histories may still contain meaningful probability mass

Hence:

$$\boxed{\frac{f_1}{N}}$$

can carry information about unseen probability mass.

## 4. Classical derivation

Assume:

$$H_1, \dots, H_N$$

are i.i.d. samples from a discrete distribution  $p(h)$ .

Pick one history  $h$ .

Let:

$$p(h) = \text{probability of observing history } h$$

We want:

$$P(C(h) = 1)$$

i.e. the probability that  $h$  appears **exactly once** in  $N$  samples.

Using the binomial model:

$$P(C(h) = 1) = \binom{N}{1} p(h) (1 - p(h))^{N-1}$$

Since:

$$\binom{N}{1} = N$$

therefore:

$$P(C(h) = 1) = Np(h)(1 - p(h))^{N-1}$$

## 5. Expected number of singleton histories

Since:

$$f_1 = \sum_h \mathbf{1}[C(h) = 1]$$

take expectations:

$$E[f_1] = \sum_h E[\mathbf{1}[C(h) = 1]]$$

For an indicator:

$$E[\mathbf{1}[A]] = P(A)$$

Therefore:

$$E[f_1] = \sum_h P(C(h) = 1)$$

Substitute:

$$E[f_1] = \sum_h Np(h)(1 - p(h))^{N-1}$$

Divide by  $N$ :

$$E\left[\frac{f_1}{N}\right] = \sum_h p(h)(1 - p(h))^{N-1}$$

## 6. Compare with expected missing mass

After  $N - 1$  samples:

$$P(h \text{ has never appeared}) = (1 - p(h))^{N-1}$$

The probability mass contributed by that unseen history is:

$$p(h)(1 - p(h))^{N-1}$$

Summing over all histories:

$$E[M_{N-1}] = \sum_h p(h)(1 - p(h))^{N-1}$$

But:

$$E\left[\frac{f_1}{N}\right] = \sum_h p(h)(1 - p(h))^{N-1}$$

Hence:

$$\boxed{E\left[\frac{f_1}{N}\right] = E[M_{N-1}]}$$

Interpretation:

singleton frequency contains information about unseen probability mass

under the classical i.i.d. sampling model.

Important:

$$E[f_1/N] = E[M_{N-1}]$$

not exactly:

$$E[M_N]$$

## 7. Why this is not automatically valid for language

For n-gram histories:

$$H_t = (w_{t-k}, \dots, w_{t-1})$$

and:

$$H_{t+1} = (w_{t-k+1}, \dots, w_t)$$

These histories overlap.

Example for trigram histories:

$$H_t = (w_{t-2}, w_{t-1})$$

$$H_{t+1} = (w_{t-1}, w_t)$$

They share:

$$w_{t-1}$$

Therefore:

$$\boxed{H_t \not\perp H_{t+1}}$$

Consecutive histories are not independent samples.

## 8. Exact step that breaks

The classical derivation used:

$$P(C(h) = 1) = Np(h)(1 - p(h))^{N-1}$$

This assumes:

- $N$  independent trials,
- the same probability  $p(h)$  on each trial,
- binomial counting.

For overlapping language histories:

independence assumption fails

Therefore:

$$P(C(h) = 1)$$

is not automatically binomial.

So we cannot automatically conclude:

$$\frac{f_1}{N} = \text{a theoretically calibrated missing-mass estimate}$$

## 9. Our terminology

For history length:

$$k = n - 1$$

define:

$$N_k = \text{total number of length-}k \text{ history occurrences}$$

$$f_1^{(k)} = \text{number of distinct length-}k \text{ histories occurring exactly once}$$

Then:

$$S_k = \frac{f_1^{(k)}}{N_k}$$

Call:

$$S_k = \text{singleton-history rate}$$

or:

Good–Turing-inspired sparsity diagnostic

Do **not** call it:

“the true missing mass”

or:

“a guaranteed unbiased missing-mass estimator”

## 10. Why $S_k$ is still useful

A high:

$$S_k$$

means many training histories are supported by only one observation.

Therefore:

$$S_k \uparrow$$

suggests:

history support becomes thinner

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probability estimates may become less reliable

But this is a hypothesis to test, not a guaranteed theorem.

## 11. First validation target: actual unseen-history rate

Define:



$$U_k = \frac{\#\{\text{held-out histories unseen in training}\}}{N_{\text{held-out},k}}$$

Then first test:

$$\boxed{S_k \longrightarrow U_k}$$

Question:

Does a higher training singleton-history rate correspond to more unseen held-out histories?

If not:

$$S_k$$

is a weak proxy for the missing-coverage idea motivating it.

## 12. Central relationship of the paper

Recall:

$$\Delta H_n = H_n - H_{n-1}$$

where:

$$\Delta H_n < 0 \Rightarrow \text{more context helped}$$

and:

$$\Delta H_n > 0 \Rightarrow \text{more context hurt}$$

Our central empirical question is:

$$\boxed{S_{n-1} \longrightarrow \Delta H_n}$$

In words:

Can training-only context sparsity predict the marginal value of more context?

### 13. Important: correlation is not enough

Suppose:

$$\rho(S, \Delta H) = 0.9$$

This does **not** automatically prove:

$$S$$

is a useful predictor.

Why?

Because:

$$n \uparrow \Rightarrow S_k \uparrow$$

and possibly:

$$n \uparrow \Rightarrow \Delta H_n \uparrow$$

So both variables may simply be changing with model order.

Training size  $m$  is also a confounder.

Therefore:

pooled correlation alone is insufficient

Later compare within fixed order transitions:

$$2 \rightarrow 3, \quad 3 \rightarrow 4, \quad 4 \rightarrow 5, \dots$$

across different training sizes.

### 14. Stage 3 summary

many singleton histories

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$S_k \uparrow$

↓

possible evidence of poor context coverage

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test whether:

$S_k$  tracks  $U_k$

then test whether:

$S_{n-1}$  tracks  $\Delta H_n$

|                                                                                  |
|----------------------------------------------------------------------------------|
| training occupancy $\longrightarrow$ possible warning of generalisation reversal |
|----------------------------------------------------------------------------------|

**Methodological decision:**  $S_k = f_1^{(k)}/N_k$  is used as a training-only singleton-history sparsity statistic motivated by Good–Turing reasoning. Because overlapping language histories violate the classical i.i.d. assumption, it is not treated as a guaranteed missing-mass estimator.